

DE-CLAWING GRAPH THEORY

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CHROMATIC POLYNOMIAL: BIRKHOFF 1912

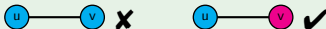
Given G with vertices $V(G)$ a **proper colouring** κ of G in k colours is

$$\kappa : V(G) \rightarrow \{1, 2, 3, \dots, k\}$$

so if $u, v \in V(G)$ are joined by an edge then

$$\kappa(u) \neq \kappa(v).$$

EXAMPLE



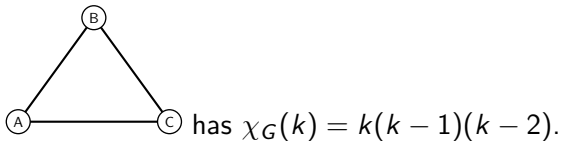
CHROMATIC POLYNOMIAL: BIRKHOFF 1912

Given G the chromatic polynomial $\chi_G(k)$ is the number of proper colourings with k colours.



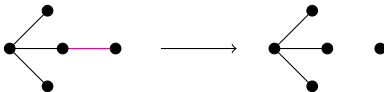
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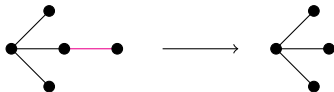


DELETION-CONTRACTION

Delete ϵ : remove edge ϵ to get $G - \epsilon$.

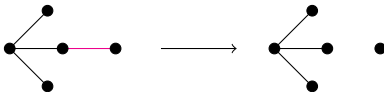


Contract ϵ : shrink edge ϵ + identify vertices to get G/ϵ .

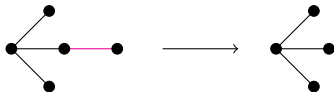


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Contract ϵ : shrink edge ϵ + identify vertices to get G/ϵ .



THEOREM (DELETION-CONTRACTION)

$$\chi_G(k) - \chi_{G-\epsilon}(k) + \chi_{G/\epsilon}(k) = 0$$

CHROMATIC SYMMETRIC FUNCTION: STANLEY 1995

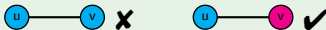
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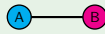


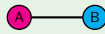
CHROMATIC SYMMETRIC FUNCTION: STANLEY 1995

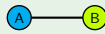
Given a proper colouring κ of vertices $v_1, v_2, \dots, v_N$ associate a monomial in commuting variables $x_1, x_2, x_3, \dots$

$$x_{\kappa(v_1)} x_{\kappa(v_2)} \cdots x_{\kappa(v_N)}.$$

EXAMPLE

 gives $x_1 x_2$.

 gives $x_2 x_1 = x_1 x_2$.

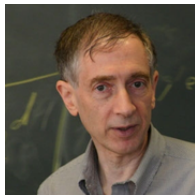
 gives $x_1 x_3$.

CHROMATIC SYMMETRIC FUNCTION: STANLEY 1995

Given G with vertices $v_1, v_2, \dots, v_N$ the chromatic symmetric function is

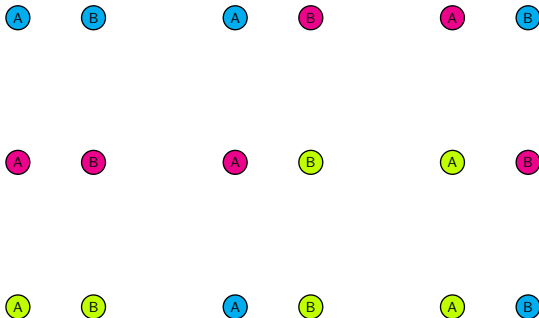
$$X_G = \sum_{\kappa} x_{\kappa(v_1)} x_{\kappa(v_2)} \cdots x_{\kappa(v_N)}$$

where the sum over all proper colourings κ .



CHROMATIC SYMMETRIC FUNCTION: STANLEY 1995

⊙ ⊙ has $X_G(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3 + 2x_1x_3$.



MULTI-DELETION

Deletion-contraction fails, as contraction gives degree change.

THEOREM (TRIPLE-DELETION: ORELLANA-SCOTT 2014)

Let G be such that $\epsilon_1, \epsilon_2, \epsilon_3$ form a triangle. Then

$$X_G - X_{G-\{\epsilon_1\}} - X_{G-\{\epsilon_2\}} + X_{G-\{\epsilon_1, \epsilon_2\}} = 0.$$

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THEOREM (k -DELETION: DAHLBERG-vW 2018)

Let G be such that $\epsilon_1, \epsilon_2, \dots, \epsilon_k$ form a k -cycle for $k \geq 3$. Then

$$\sum_{S \subseteq [k-1]} (-1)^{|S|} X_{G-\cup_{i \in S} \{\epsilon_i\}} = 0.$$

(Crew-Spirkl 2020) Deletion-contraction for vertex-weighted chromatic symmetric function.

SYMMETRIC FUNCTIONS

A **symmetric function** is a formal power series f in commuting variables $x_1, x_2, \dots$ such that for all permutations π

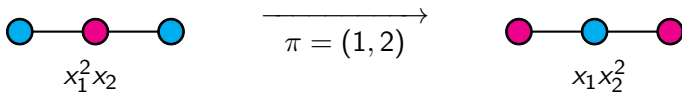
$$f(x_1, x_2, \dots) = f(x_{\pi(1)}, x_{\pi(2)}, \dots).$$

SYMMETRIC FUNCTIONS

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$$f(x_1, x_2, \dots) = f(x_{\pi(1)}, x_{\pi(2)}, \dots).$$

X_G is a symmetric function.



Let

$$\Lambda = \bigoplus_{N \geq 0} \Lambda^N \subset \mathbb{Q}[[x_1, x_2, \dots]]$$

be the **algebra of symmetric functions** with Λ^N spanned by ...

BASIS: POWER SUM

A **partition** $\lambda = \lambda_1 \geq \cdots \geq \lambda_\ell > 0$ of N is a list of positive integers whose sum is N : **3221** $\vdash$ **8**.

The i -th power sum symmetric function is

$$p_i = x_1^i + x_2^i + x_3^i + \cdots$$

and for $\lambda = \lambda_1 \cdots \lambda_\ell$

$$p_\lambda = p_{\lambda_1} \cdots p_{\lambda_\ell}.$$

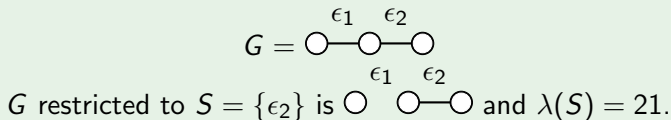
EXAMPLE

$$p_{21} = p_2 p_1 = (x_1^2 + x_2^2 + x_3^2 + \cdots)(x_1 + x_2 + x_3 + \cdots)$$

BASIS: POWER SUM

Given $S \subseteq E(G)$, $\lambda(S)$ is the partition determined by the connected components of G restricted to S .

EXAMPLE



THEOREM (STANLEY 1995)

$$X_G = \sum_{S \subseteq E(G)} (-1)^{|S|} p_{\lambda(S)}$$

BASIS: POWER SUM

$$G = \overset{\epsilon_1}{\bigcirc} - \overset{\epsilon_2}{\bigcirc} - \bigcirc$$

G restricted to

- $S = \{\epsilon_1, \epsilon_2\}$ is $\overset{\epsilon_1}{\bigcirc} - \overset{\epsilon_2}{\bigcirc} - \bigcirc$ and $\lambda(S) = 3$
- $S = \{\epsilon_1\}$ is $\overset{\epsilon_1}{\bigcirc} - \overset{\epsilon_2}{\bigcirc} \quad \bigcirc$ and $\lambda(S) = 21$
- $S = \{\epsilon_2\}$ is $\overset{\epsilon_1}{\bigcirc} \quad \overset{\epsilon_2}{\bigcirc} - \bigcirc$ and $\lambda(S) = 21$
- $S = \emptyset$ is $\overset{\epsilon_1}{\bigcirc} \quad \overset{\epsilon_2}{\bigcirc} \quad \bigcirc$ and $\lambda(S) = 111$.

$$X_G = p_3 - 2p_{21} + p_{111}$$

BASIS: ELEMENTARY

The i -th elementary symmetric function is

$$e_i = \sum_{j_1 < \dots < j_i} x_{j_1} \cdots x_{j_i}$$

and for $\lambda = \lambda_1 \cdots \lambda_\ell$

$$e_\lambda = e_{\lambda_1} \cdots e_{\lambda_\ell}.$$

EXAMPLE

$$e_{21} = e_2 e_1 = (x_1 x_2 + x_1 x_3 + x_2 x_3 + \cdots)(x_1 + x_2 + x_3 + \cdots)$$

BASIS: ELEMENTARY

THEOREM (STANLEY 1995)

If

$$X_G = \sum_{\lambda} c_{\lambda} e_{\lambda}$$

then

$$\sum_{\lambda \text{ with } k \text{ parts}} c_{\lambda} = \text{number of acyclic orientations with } k \text{ sinks.}$$

EXAMPLE



$$X_G = 3e_3 + e_{21}$$

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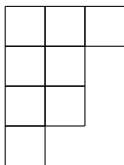


$$X_G = 3e_3 + e_{21}$$

PARTITIONS AND DIAGRAMS

A **partition** $\lambda = \lambda_1 \geq \cdots \geq \lambda_\ell > 0$ of N is a list of positive integers whose sum is N : **3221** $\vdash$ 8.

The **diagram** $\lambda = \lambda_1 \geq \cdots \geq \lambda_\ell > 0$ is the array of **boxes** with λ_i boxes in row i from the **top**.



3221

SEMI-STANDARD YOUNG TABLEAUX

A semi-standard Young tableau (SSYT) T of shape λ is a filling with $1, 2, 3, \dots$ so rows **weakly increase** and columns **increase**.

1	1	1
2	4	
4	5	
6		

Given an SSYT T we have

$$x^T = x_1^{\#1s} x_2^{\#2s} x_3^{\#3s} \dots$$

$$x_1^3 x_2 x_4^2 x_5 x_6$$

BASIS: SCHUR

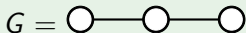
The Schur function is

$$s_{\lambda} = \sum_{T \text{ SSYT of shape } \lambda} x^T.$$

EXAMPLE

$$s_{21} = x_1^2 x_2 + x_1 x_2^2 + x_1^2 x_3 + x_1 x_3^2 + x_2^2 x_3 + x_2 x_3^2 + 2x_1 x_2 x_3 + \cdots$$

1	1	1	2	1	1	1	3	2	2	2	3	1	2	1	3
2		2		3		3		3		3		3		2	



$$X_G = s_{21} + 4s_{111}$$

(Wang-Wang 2020) Intricate formula for X_G .

e-POSITIVITY AND SCHUR-POSITIVITY

G is **e-positive** if X_G is a positive linear combination of e_λ .

G is **Schur-positive** if X_G is a positive linear combination of s_λ .

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 has $X_G = e_{211} - 2e_{22} + 5e_{31} + 4e_4$ ✗
 $X_G = 8s_{1111} + 5s_{211} - s_{22} + s_{31}$ ✗

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has

$$\begin{aligned} X_G &= e_{21} + 3e_3 \quad \checkmark \\ X_G &= 4s_{111} + s_{21} \quad \checkmark \end{aligned}$$



has

$$\begin{aligned} X_G &= e_{211} - 2e_{22} + 5e_{31} + 4e_4 \quad \times \\ X_G &= 8s_{1111} + 5s_{211} - s_{22} + s_{31} \quad \times \end{aligned}$$

Claw aka $K_{1,3}$: Smallest graph that is not e-positive. Smallest graph that is not Schur-positive. Central to positivity.

e-POSITIVITY AND SCHUR-POSITIVITY

For $\lambda = \lambda_1 \cdots \lambda_\ell$

$$e_\lambda = \sum_{\mu} K_{\mu\lambda} s_{\mu^t}$$

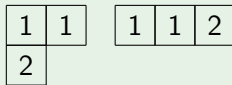
where $K_{\mu\lambda} = \#$ SSYTs of shape μ filled with λ_1 1s, $\dots$, λ_ℓ ℓ s, and μ^t is the transpose of μ along the downward diagonal.

Hence $K_{\mu\lambda} \geq 0$ and

e-positivity implies Schur-positivity.

EXAMPLE

$$e_{21} = s_{21} + s_{111}$$

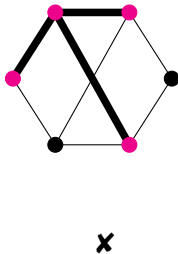
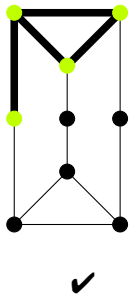


WHY e -POSITIVITY AND SCHUR-POSITIVITY?

- If e -positive, then it is related to permutation representations.
- We have e -positivity implies Schur-positivity.
- If Schur-positive, then it arises as the Frobenius image of some representation of a symmetric group.
- If Schur-positive, then it arises as the character of a polynomial representation of a general linear group.
- The Gasharov-Stanley claw-free problem.
- The Stanley-Stembridge problem. Proof using probability by Hikita (2024); LLT polynomials by Griffin, Mellit, Romero, Weigl and Wen (2025).

CLAW-FREE GRAPHS

Claw-free: does not contain the claw as an induced subgraph of the graph.



e-POSITIVITY AND SCHUR-POSITIVITY

THEOREM (GASHAROV 1996)

If G is a *claw-free incomparability graph* then G is Schur-positive.

Raised whether all claw-free graphs are Schur-positive.

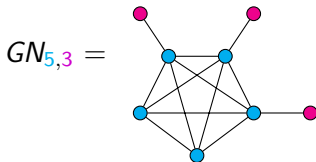
PROBLEM (GASHAROV-STANLEY 1996)

If G is a claw-free graph then G is Schur-positive.

True for $N \leq 9$ vertices.

e-POSITIVITY AND SCHUR-POSITIVITY

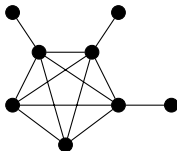
The generalized net $GN_{n,m}$ for $n \geq 1, n \geq m \geq 0$ is given by the complete graph K_n with m degree 1 vertices attached *distinctly*.



THEOREM (SHELBURNE-VW 2025)

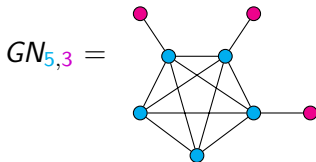
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CF:



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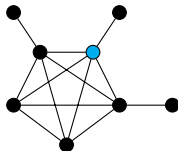
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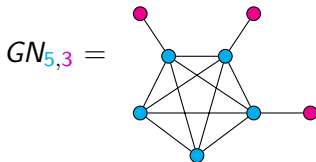
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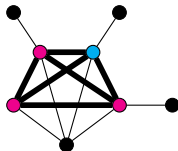
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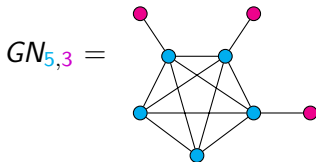
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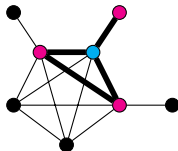
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If G is a *claw-free incomparability graph* then G is Schur-positive.

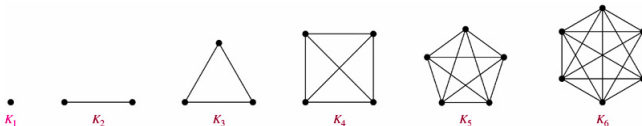
A large special case of the...

PROBLEM (STANLEY-STEMBRIDGE 1993)

If G is a *claw-free incomparability graph* then G is e-positive.

e-POSITIVITY AND SCHUR-POSITIVITY

Guay-Paquet showed enough to prove it for **unit interval graphs**, namely a **connected intersection of complete graphs** in a row.



PROBLEM (STANLEY-STEMBRIDGE 1993)

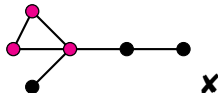
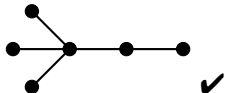
If G is a connected intersection of complete graphs then G is e-positive.

EXAMPLE



WIDEN THE PROBLEM - PART 1

e-POSITIVITY OF TREES: DAHLBERG-SHE-VW 2020



N	1	2	3	4	5	6	7	8	9	10	11	12	13
trees	1	1	1	2	3	6	11	23	47	106	235	551	1301
e-pos	1	1	1	1	2	1	3	1	2	2	5	1	4

e-POSITIVITY OF TREES

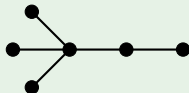
THEOREM (DAHLBERG-SHE-VW 2020)

Any tree with N vertices and a vertex of degree

$$d \geq \log_2 N + 1$$

*is **not** e-positive.*

EXAMPLE



is **not** e-positive.

e-POSITIVITY OF TREES

CONJECTURE (DAHLBERG-SHE-VW 2020)

Any tree with N vertices and a vertex of degree

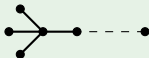
$$d \geq 4$$

*is **not** e-positive.*

(Zheng 2020) True for $d \geq 6$.

(Tom 2024) True for $d = 5$, and $d = 4$ not adjacent to leaf.

EXAMPLE

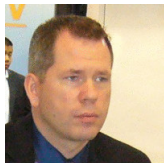


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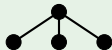
e-POSITIVITY TEST OF WOLFGANG III 1997

A graph has a **connected partition** of type $\lambda = \lambda_1 \cdots \lambda_\ell$ if we can find disjoint subsets of vertices $V_1, \dots, V_\ell \in V(G)$ so

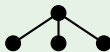
- $V_1 \cup \dots \cup V_\ell = V(G)$
- restricting edges to each V_i gives connected components with λ_i vertices.



EXAMPLE



has connected partitions of type 4, 31, 211 and 1111



but is missing a connected partition of type 22.

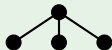
e-POSITIVITY TEST OF WOLFGANG III 1997

THEOREM (WOLFGANG III 1997)

If a connected graph G with N vertices is e-positive, then G has a connected partition of type λ for every partition $\lambda \vdash N$.

Test: If G does **not** have a connected partition of some type then G is **not** e-positive.

EXAMPLE



does **not** have a connected partition of type 22 . Hence it is **not** e-positive.

SCHUR-POSITIVITY OF TREES

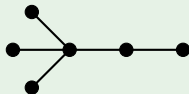
THEOREM (DAHLBERG-SHE-VW 2020)

Any tree with N vertices and a vertex of degree

$$d > \left\lceil \frac{N}{2} \right\rceil$$

*is **not** Schur-positive.*

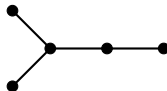
EXAMPLE



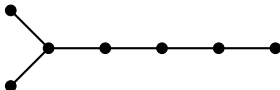
is **not** Schur-positive.

PROBLEMS

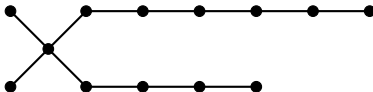
- ① Which trees with max degree 3 are e-positive?



- ② Which trees are Schur-positive but not e-positive?



- ③ Which trees pass Wolfgang's test but are not e-positive?



WIDEN THE PROBLEM - PART 2

STANLEY'S WIDENING: DAHLBERG-FOLEY-VW

ROBBINS PRIZE 2023

Stanley 1995:

We don't know of a graph which is not contractible to $K_{1,3}$ (even regarding multiple edges of a contraction as a single edge) which is not e-positive.

WIDEN THE PROBLEM - PART 2

STANLEY'S WIDENING: DAHLBERG-FOLEY-VW ROBBINS PRIZE 2023

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We don't know of a graph which is not contractible to $K_{1,3}$ (even regarding multiple edges of a contraction as a single edge) which is not e-positive.

We do.



THE DAVID P. ROBBINS PRIZE, OF THE MATHEMATICAL ASSOCIATION OF AMERICA

“The David P. Robbins Prize of \$5000 is awarded every third year at a national meeting of the Association. Papers are judged on quality of research, clarity of exposition, and accessibility to undergraduates.”

Since 1995 inaugural award

- No Canadians had won the prize.
- No women had won the prize.



WHAT LED TO OUR SUCCESS?



Our definition of success:

- Enjoy the problem;
- Enjoy the solution;
- Enjoy the collaboration.

And celebrate if others enjoy your results too!

WHAT LED TO OUR SUCCESS?



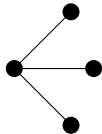
Our definition of success:

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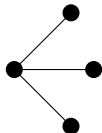
And celebrate if others enjoy your results too!

The only thing you control in life is your reactions!

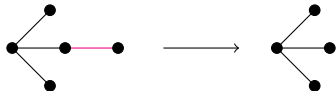
THE CLAW AKA $K_{1,3}$



THE CLAW AKA $K_{1,3}$



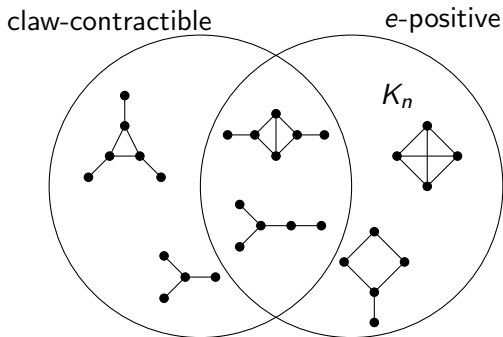
Contracts to the claw: shrinking edges + identifying vertices + removing multiple edges = claw.



A PICTURE SPEAKS 1000 WORDS

Stanley 1995:

We don't know of a graph which is not contractible to $K_{1,3}$ (even regarding multiple edges of a contraction as a single edge) which is not e-positive.



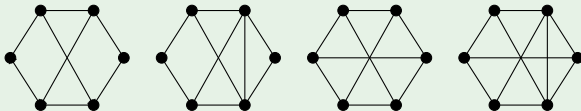
CLAW-CONTRACTIBLE-FREE: BROUWER-VELDMAN 1987

Delete v : remove vertex v and attached edges to get $G - v$.



G is **claw-contractible-free** if and only if deleting all sets of 3 **non-adjacent** vertices gives disconnection.

EXAMPLE



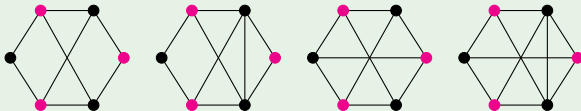
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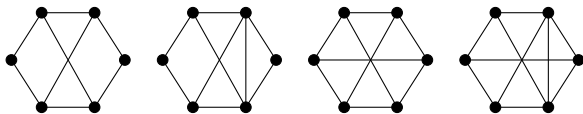


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EXAMPLE

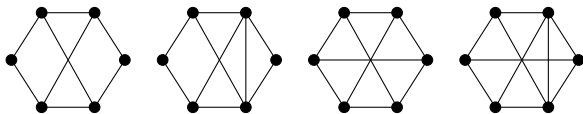


...WITH CHROMATIC SYMMETRIC FUNCTION



$$\begin{array}{rclclclcl}
 2e_{222} & - & 6e_{33} & + & 26e_{42} & + & 28e_{51} & + & 102e_6 \\
 2e_{321} & - & 6e_{33} & + & 24e_{42} & + & 40e_{51} & + & 120e_6 \\
 2e_{222} & - & 12e_{33} & + & 30e_{42} & + & 24e_{51} & + & 186e_6 \\
 2e_{321} & - & 6e_{33} & + & 20e_{42} & + & 32e_{51} & + & 228e_6
 \end{array}$$

...WITH CHROMATIC SYMMETRIC FUNCTION

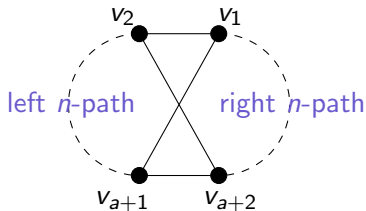
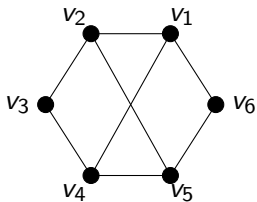


$$\begin{array}{rclclcl}
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 2e_{321} & - & 6e_{33} & + & 20e_{42} & + & 32e_{51} & + & 228e_6
 \end{array}$$

Smallest counterexamples to Stanley's statement.

INFINITE FAMILY: SALTIRE GRAPHS

The saltire graph $SA_{n,n}$ for $n \geq 3$ is given by



with $SA_{3,3}$ on the left.

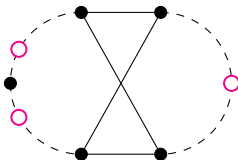
INFINITE FAMILY: SALTIRE GRAPHS

THEOREM (DAHLBERG-FOLEY-vW 2020)

$SA_{n,n}$ for $n \geq 3$ is claw-contractible-free and

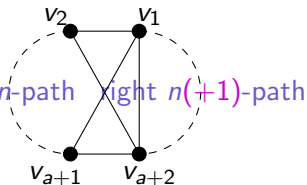
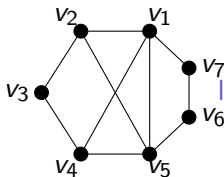
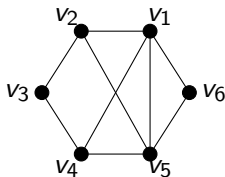
$$[e_{nn}]X_{SA_{n,n}} = -n(n-1)(n-2).$$

CCF:



FOR ANY n : AUGMENTED SALTIRE GRAPHS

The augmented saltire graphs $AS_{n,n}$, $AS_{n,n+1}$ for $n \geq 3$.



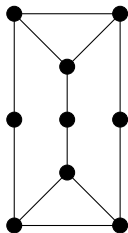
THEOREM (DAHLBERG-FOLEY-vW 2020)

$AS_{n,n}$ and $AS_{n,n+1}$ for $n \geq 3$ are claw-contractible-free and

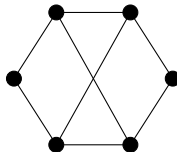
$$[e_{nn}]X_{AS_{n,n}} = [e_{(n+1)n}]X_{AS_{n,n+1}} = -n(n-1)(n-2).$$

CLAW-FREE: BEINEKE 1970

Claw-free: does not contain the claw as an induced subgraph of the graph.



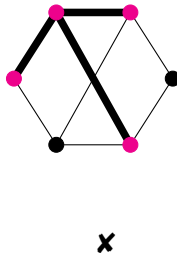
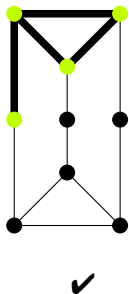
✓



✗

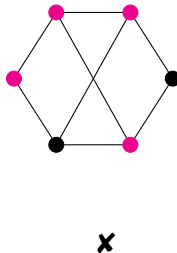
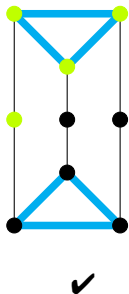
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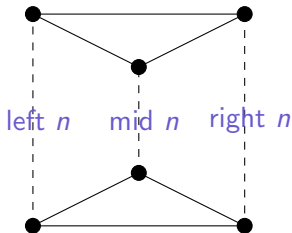
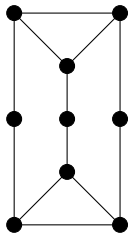
CLAW-FREE: BEINEKE 1970

G is **claw-free** if there exists an edge partition giving complete graphs, every vertex in at most two.



AND CLAW-FREE: TRIANGULAR TOWER GRAPHS

The triangular tower graph $TT_{n,n,n}$ for $n \geq 3$ is given by



with $TT_{3,3,3}$ on the left.

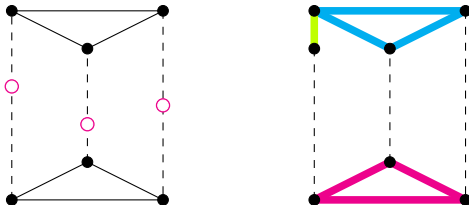
AND CLAW-FREE: TRIANGULAR TOWER GRAPHS

THEOREM (DAHLBERG-FOLEY-vW 2020)

$TT_{n,n,n}$ for $n \geq 3$ is claw-contractible-free, claw-free and

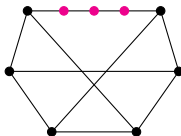
$$[e_{nnn}]X_{TT_{n,n,n}} = -n(n-1)^2(n-2).$$

CCF+CF:



CONJECTURES

- ① Bloated $K_{3,3}$:



with $3n$ vertices has

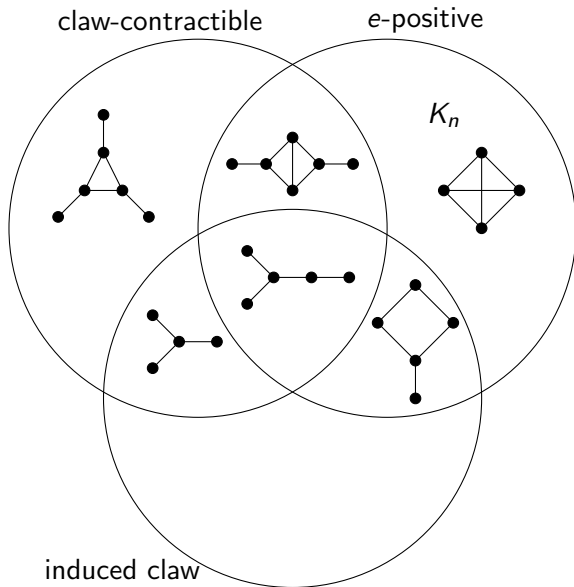
$$-(3 \times 2^n)e_{3^n}.$$

- ② No G exists that is connected, claw-contractible-free, claw-free and not e-positive with 10, 11 vertices

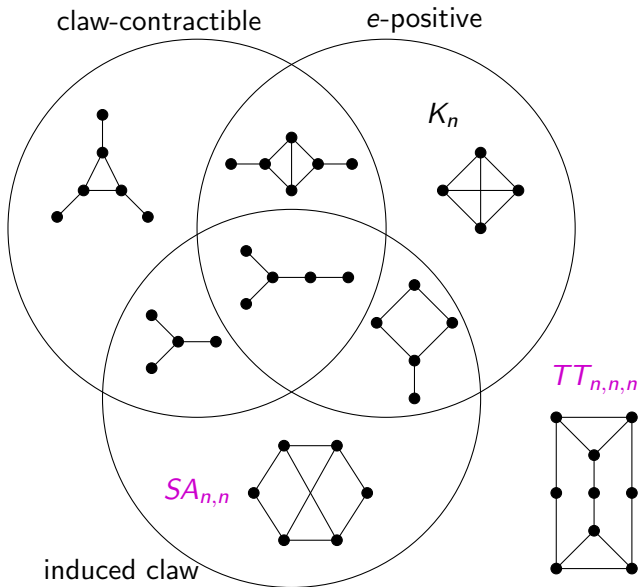
SCARCITY

- $N = 6$: 4 of 112 connected graphs ccf and not e -positive.
- $N = 7$: 7 of 853 connected graphs ccf and not e -positive.
- $N = 8$: 27 of 11117 connected graphs ccf and not e -positive.
- Of 293 terms in $TT_{7,7,7}$ only $-ve$ at e_{777} .
- Of 564 terms in $TT_{8,8,8}$ only $-ves$ at e_{888} and $-1944e_{4444444}$.
- Of 1042 terms in $TT_{9,9,9}$ only $-ves$ at e_{999} , $-768e_{333333333}$.

A PICTURE SPEAKS 1000 WORDS



A PICTURE SPEAKS 1000 WORDS



In general, e-positivity has nothing to do with the claw.



Thank you very much!